Unit: 6 Unit 8: Quadratics
Topic: Vertex Form and Intercept Form

Name: ____________
Date: ________

Challenge (Habanero)

Q1. A chef is designing a kitchen with a quadratic-shaped counter. The vertex form of the counter's shape is $y = a(x-h)^2 + k$. If $a = 2$, $h = 3$, and $k = -1$, what is the equation of the counter's shape?
(A) $y = 2(x-3)^2 - 1$
(B) $y = 2(x+3)^2 + 1$
(C) $y = 2(x-3)^2 + 1$
(D) $y = 2(x+3)^2 - 1$
(E) $y = 2(x-2)^2 - 1$

Q2. A builder is constructing a quadratic-shaped roof. The intercept form of the roof's shape is $y = a(x-p)(x-q)$. If $p = 2$ and $q = 5$, what is the value of a if the roof passes through the point $(3, 12)$?
(A) $a = 2$
(B) $a = 3$
(C) $a = 4$
(D) $a = 5$
(E) $a = 6$

Q3. A recipe for a quadratic-shaped cake requires x cups of flour and y cups of sugar. The vertex form of the cake's shape is $y = a(x-h)^2 + k$. If $a = -3$, $h = 2$, and $k = 10$, what is the maximum amount of sugar required?
(A) $y = 4$
(B) $y = 6$
(C) $y = 8$
(D) $y = 10$
(E) $y = 12$

Q4. An architect is designing a quadratic-shaped bridge. The intercept form of the bridge's shape is $y = a(x-p)(x-q)$. If $a = 1$, $p = -2$, and $q = 4$, what are the x-intercepts of the bridge?
(A) $x = -2, x = 4$
(B) $x = -4, x = 2$
(C) $x = -1, x = 3$
(D) $x = 1, x = 5$
(E) $x = -3, x = 1$

Q5. A construction team is building a quadratic-shaped tunnel. The vertex form of the tunnel's shape is $y = a(x-h)^2 + k$. If $a = -2$, $h = 1$, and $k = 5$, what is the equation of the tunnel's shape?
(A) $y = -2(x-1)^2 + 5$
(B) $y = -2(x+1)^2 - 5$
(C) $y = -2(x-1)^2 - 5$
(D) $y = -2(x+1)^2 + 5$
(E) $y = -2(x-2)^2 + 5$

Q6. A chef is making a quadratic-shaped pastry. The intercept form of the pastry's shape is $y = a(x-p)(x-q)$. If $p = 1$ and $q = 3$, what is the value of a if the pastry passes through the point $(2, 6)$?
(A) $a = 3$
(B) $a = 4$
(C) $a = 5$
(D) $a = 6$
(E) $a = 7$

Q7. An engineer is designing a quadratic-shaped roller coaster track. The vertex form of the track's shape is $y = a(x-h)^2 + k$. If $a = 1$, $h = -2$, and $k = 3$, what is the minimum height of the track?
(A) $y = 1$
(B) $y = 2$
(C) $y = 3$
(D) $y = 4$
(E) $y = 5$

Q8. A builder is constructing a quadratic-shaped foundation. The intercept form of the foundation's shape is $y = a(x-p)(x-q)$. If $a = -1$, $p = -1$, and $q = 2$, what are the x-intercepts of the foundation?
(A) $x = -1, x = 2$
(B) $x = -2, x = 1$
(C) $x = -3, x = 1$
(D) $x = 1, x = 3$
(E) $x = -2, x = 3$

Open Ended Questions (FRQ)

Q9. A quadratic-shaped fountain has the equation $y = a(x-h)^2 + k$. If $a = 2$, $h = 1$, and the fountain passes through the point $(3, 12)$, find the value of k and the equation of the fountain's shape. Show all work and explain your reasoning.

Q10. A chef is designing a quadratic-shaped cookie cutter. The intercept form of the cookie cutter's shape is $y = a(x-p)(x-q)$. If $p = 2$ and $q = 4$, and the cookie cutter passes through the point $(3, 6)$, find the value of a and the equation of the cookie cutter's shape. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Remind students to identify the values of a, h, and k in vertex form and a, p, and q in intercept form.
- Tip 2: Encourage students to use the given points to find the value of a in intercept form.
- Tip 3: Emphasize the importance of showing all work and explaining reasoning in free-response questions.